

ST. MARY'S UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE



JobPortal - Mobile Application Documentation

BY

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1. Introduction

1.1 Overview

JobPortal is a comprehensive mobile application designed to connect job seekers with employers. The app provides a platform where employers can post job openings, and job seekers can browse, save, and apply for jobs. The application includes role-based access control with three user types: Job Seeker, Employer, and Administrator.

1.2 Objective of the Project

The purpose of this application is to:

- Provide an easy-to-use platform for job searching and recruitment
- Enable employers to reach potential candidates efficiently
- Allow administrators to manage users and content
- Store data locally using the Room database for offline access

1.3 Target Audience

User Type	Description
Job Seekers	Individuals looking for employment opportunities
Employers	Companies or individuals posting job vacancies
Administrators	System managers overseeing platform operations

2. Key Features

2.1 Authentication System

- User Registration (Seeker/Employer)
- User Login with email/password
- Role-based navigation

2.2 Job Seeker Features

- Browse all available jobs
- Search jobs by title or company
- Save jobs to favorites
- Apply for jobs with a cover letter and CV
- View saved jobs
- Change password
- Help section with admin contact

2.3 Employer Features

- Post new job listings
- View only their own job postings
- Edit job details
- Delete job postings
- Close jobs (marks as "CLOSED")
- Change password

2.4 Admin Features

- View all users (Seekers & Employers)
- View all job postings
- View all applications
- Delete any user
- Delete any job
- Promote users to admin
- Demote admin users
- Change password
- System statistics dashboard

2.5 General Features

- Change password for all users
- Help menu with admin contact
- Logout functionality
- Search/filter jobs

3. Tools & Environment

3.1 Development Environment

Tool	Version
Android Studio Panda	2023.1.1
Java 17	17
Gradle 8.0	8
Minimum SDK	API 24 (Android 7.0)
Target SDK	API 34 (Android 14

3.2 Dependencies/Libraries

Library	Purpose
Room	Local database management
RecyclerView	Display scrollable lists
CardView	Styled job cards
Material Design Components	UI components
ViewPager2	Tab navigation in Admin
Activity Result API	File picker for CV upload

3.3 Dependencies/Libraries

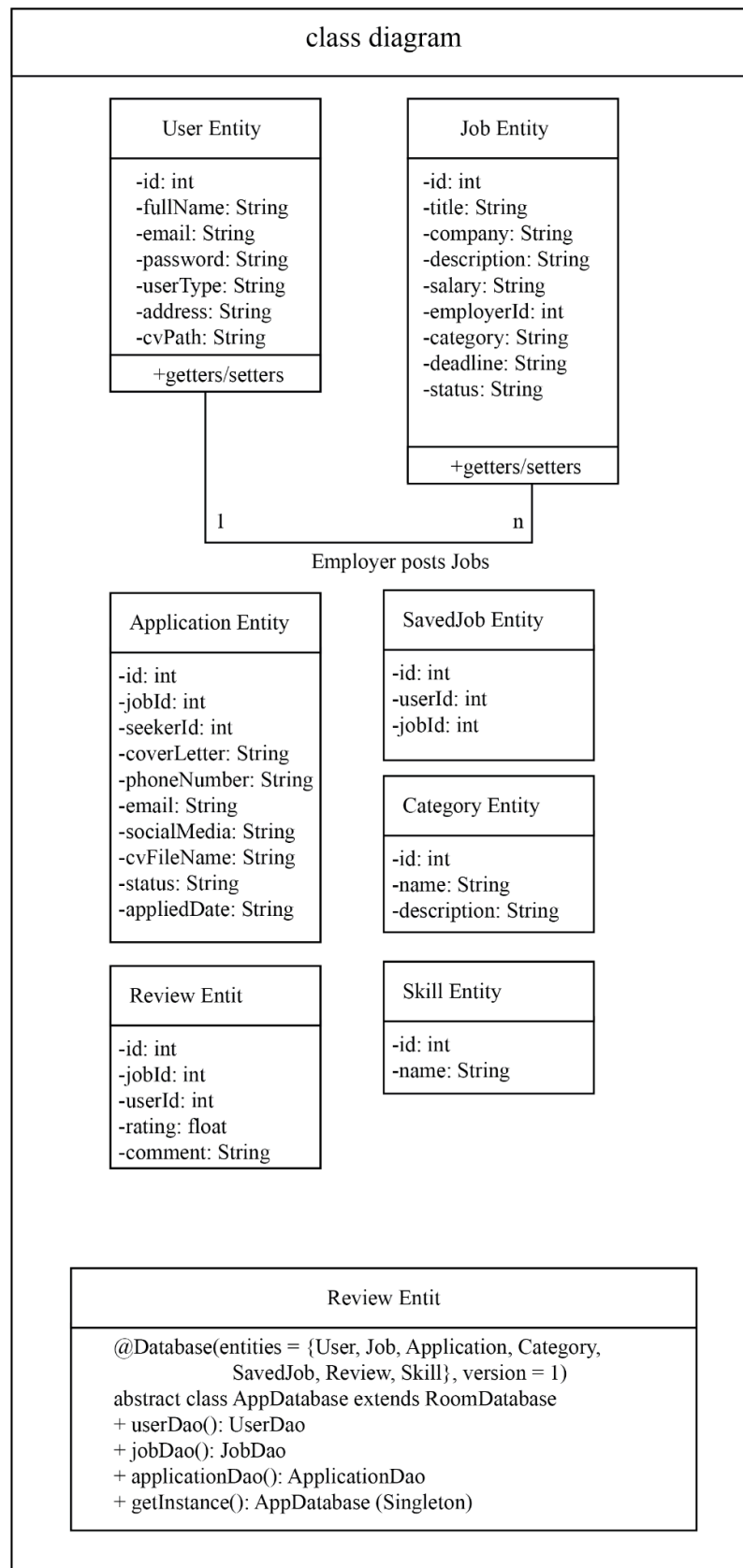
- RAM: 8GB minimum (16GB recommended)
- Storage: 500MB free space
- Android device with API 24+

4. System Design

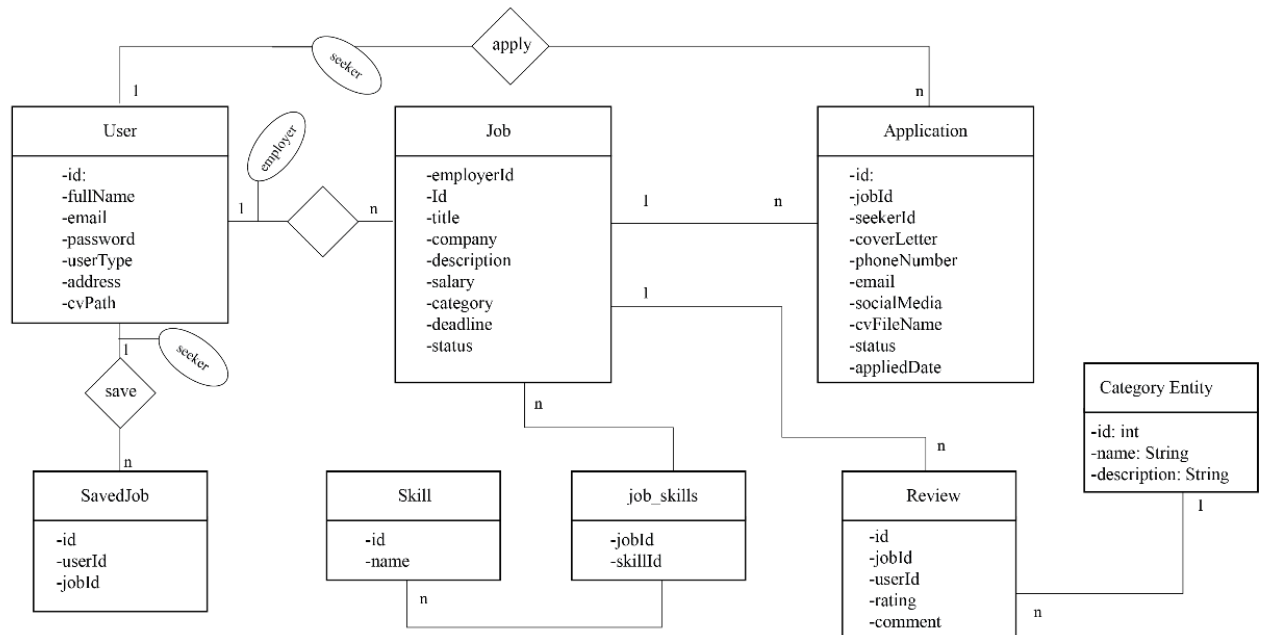
4.1 Actors & Roles

Actor	Permissions
Job Seeker	View jobs, Save jobs, Apply for jobs, Change password
Employer	Post jobs, Edit/Delete own jobs, Close jobs
Admin	Full CRUD on users, jobs, applications; Promote/Demote users

4.3 Class Diagram



5 ER Diagram



6 Functional Requirements (CRUD)

6.1 CRUD Summary Table

Entity	Create (C)	Read (R)	Update (U)	Delete (D)
User	Register	Login, Profile View	Edit Profile, Change Password	Admin on
Job	Post Job	View Jobs, Search	Edit Job (Employer), Close Job	Employer/Admin
Application	Apply	View Applications	Update Status	Admin only
SavedJob	Save Job	View Saved Jobs	-	Unsave Jo
Review	-	-	-	-
Category	-	-	-	-
Skill	-	-	-	-

6.2 Detailed CRUD Operations

User Management

- ****Create****: Registration with email, password, user type
- ****Read****: Login authentication, profile viewing
- ****Update****: Change password, edit profile (name, address, CV)
- ****Delete****: Admin can delete any user

Job Management

- ****Create****: Employers post jobs (title, company, salary, description, deadline)
- ****Read****: All users can view jobs; employers see only their own
- ****Update****: Employers can edit/close their jobs
- ****Delete****: Employers can delete own jobs; Admin can delete any job

Application Management

- ****Create****: Seekers apply with cover letter, phone, CV
- ****Read****: Seekers view their applications; Employers view applicants
- ****Update****: Admin can update application status
- ****Delete****: Admin can delete applications

Saved Job

- ****Create****: Seekers can save jobs
- ****Read****: Seekers can view saved jobs list
- ****Update****: N/A
- ****Delete****: Seekers can unsave jobs

7 Conclusion

7.1 Project Summary

The JobPortal mobile application successfully implements a complete job marketplace with three distinct user roles:

Completed Features

- User authentication (Register/Login)
- Role-based access control (Seeker/Employer/Admin)
- Job posting and management
- Job browsing with search/filter
- Save jobs functionality
- Job application with CV upload
- Admin dashboard for moderation
- Change password for all users

- Help section with admin contact

7.2 Technical Achievements

- Implemented Room Database with 7 tables
- Used MVVM architecture pattern
- Thread-safe database operations
- RecyclerView for efficient list display
- Material Design UI components

7.3 Future Enhancements

- Push notifications for new jobs
- In-app messaging between employers and seekers
- Resume builder feature
- Advanced job recommendations using AI
- Video interview scheduling
- Multi-language support

7.4 Conclusion

The JobPortal app demonstrates a complete Android application covering all core concepts: Activities, RecyclerView, Room Database, background threading, and role-based UI. The app is ready for deployment and can be extended with additional features.